INCH-POUND

MIL-C-82813(OS)
29 December 1989
SUPERSEDING
WS 20648A
15 July 1986

MILITARY SPECIFICATION

CURING AGENT, ISOPHORONE DIISOCYANATE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers the requirements for an isophorone diisocyanate curing agent, referred to herein as IPDI.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 203 Water Using Karl Fischer Reagent (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the IPDI shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 **Physical and chemical properties.** Physical and chemical properties of the IPDI shall conform to table I.

TABLE I. Physical and chemical properties.

Property	Limits	
	Minimum	Maximum
Isocyanate, equivalent/100g	0.885	0.905
Moisture, percent	---	0.02
Specific gravity, 25°C/25°C	1.050	1.070
Viscosity, centipoise	9.89	12.9
Iron, percent	---	0.002
Hydrolyzable chloride, percent	---	0.02

3.3 **Infrared (IR) spectrum.** The infrared spectrum of the IPDI shall compare qualitatively to figure 1.

3.4 **Shelf life.** The IPDI shall have a shelf life of 12 months after date of manufacture when stored at 80 ± 20°F in the original, closed containers.

3.4.1 **Shelf life extension.** The shelf life expiration date of an individual lot may be extended for additional 12-month periods provided the IPDI, upon retest, conforms to the isocyanate and moisture requirements of 3.2.

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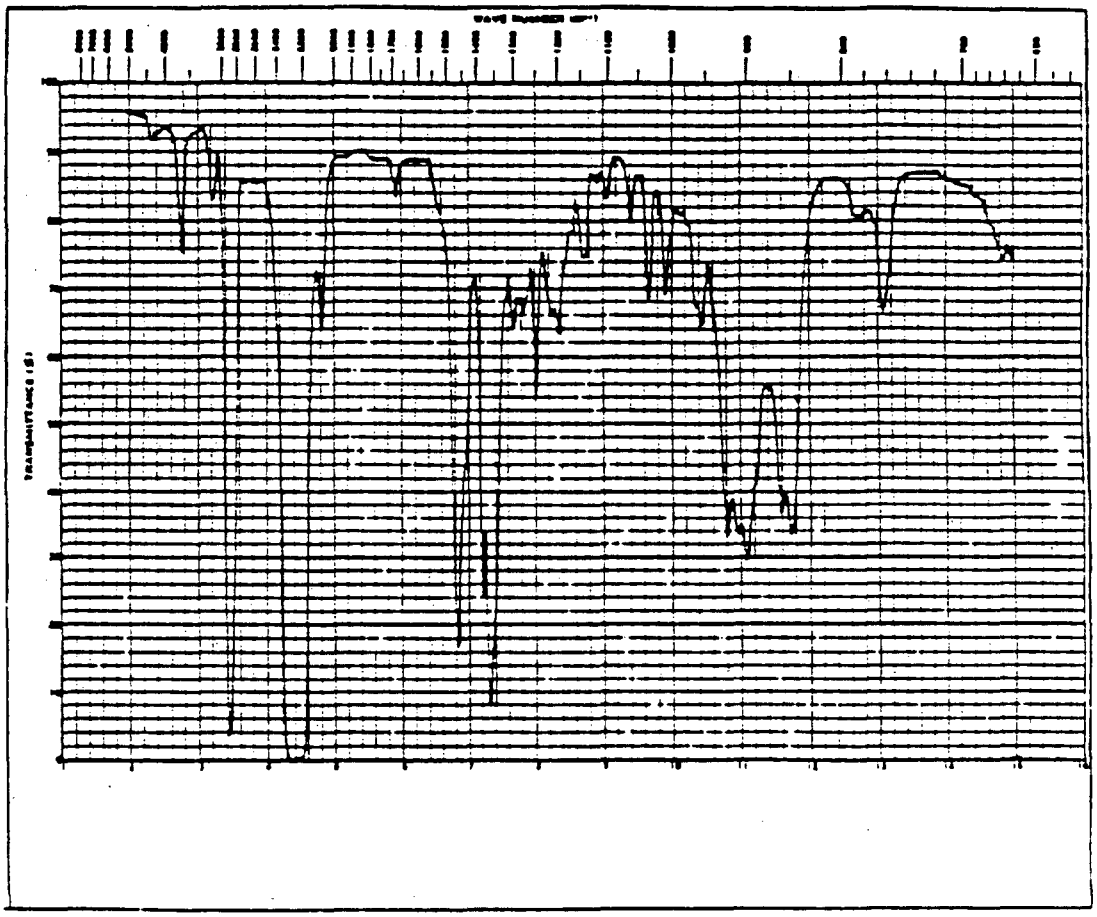


FIGURE 1. Infrared spectrum.

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3.5 **Safety.**

3.5.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.5.2 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit an MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

3.6 **Workmanship.** Workmanship shall be such that the IPDI is uniform in appearance and free from visible contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in table II.

TABLE II. Quality conformance inspection.

Tests and Examinations	Requirement	Test Method
Visual examination	3.6 and Section 5	4.5.1
Isocyanate content	3.2	4.5.2
Moisture	3.2	4.5.3
Specific gravity	3.2	4.5.4
Viscosity	3.2	4.5.5
Iron	3.2	4.5.6
Hydrolyzable chloride	3.2	4.5.7
IR identification	3.3	4.5.8

4.4.1 **Lot.** A lot shall consist of all the IPDI subject to inspection at the same time and manufactured by identical processes from one of the following:

a. A single batch (see 6.3.1).

b. A continuous process (see 6.3.2).

c. A uniform blend of batches or a uniform blend of portions from a continuous process.

4.4.2 **Sampling.** Sample size shall be in accordance with table III. The single samples from a lot shall be combined and blended to form three composite samples. An individual single sample shall not be split between composites, except where a lot consists of fewer than three containers. A single determination of each quality characteristic listed in table I shall be performed on each of the three composite samples independently. When a lot consists of fewer than three containers, three samples shall be taken from each container. The sample portions thus obtained shall be grouped and blended to form three composite samples. The results of each test shall conform to the requirements of table I. Failure of any composite sample to meet any requirement shall be cause for rejection of the lot.

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TABLE III. Sample size.

Number of containers in lot	Number of containers to be selected
1 - 3	All
4 - 40	3
41 - 65	4
66 - 110	5
111 - 500	7

4.5 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Visual examination shall be made to verify conformance to the requirements of 3.6 and section 5.

4.5.2 **Isocyanate content.** The isocyanate content shall be determined in accordance with the following:

a. Prepare a solution of 0.1N di-n-butylamine in dry toluene. (Note: "dry" is defined as containing less than 0.001 percent water.) Allow to stand 24 hours minimum before using.

b. Weigh, to the nearest 0.1 mg, 0.25 to 0.30 g of sample into a dry flask.

c. Using a pipette, add exactly 40 ml of 0.1N di-n-butylamine solution. Stopper flask and swirl. Allow to stand for 5 minutes at room temperature.

d. Add a few drops of bromocresol green indicator solution, and titrate with standard 0.1N methanolic hydrochloric acid (HCl) to yellow endpoint which persists for at least 15 seconds.

e. Run at least two blank determinations using same solutions and in same manner as above but omitting the sample.

f. Calculate isocyanate content as follows:

$$\text{Isocyanate, eq/100g} = \frac{N\ (B - A)}{10\ W}$$

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Where:

B = volume of standard HCl for blank, ml
A = volume of standard HCl for sample, ml
N = normality of standard HCl
W = sample weight, g
10 = standard factor to convert milliequivalent/g to equivalent/100 g

g. Record isocyanate content to the nearest 0.001 eq/100 g

4.5.3 **Moisture.** Moisture shall be determined in accordance with ASTM E 203, using a 9- to 10-g sample, and titrating to a 30-second endpoint using a carrier solution of 1:1 pyridine chloroform. Moisture shall be recorded to the nearest 0.01 percent.

4.5.4 **Specific gravity.** Specific gravity shall be determined in accordance with the following:

a. Accurately weigh a calibrated, clean, dry pycnometer.

b. Slightly overfill the pycnometer with sample, keeping tightly trapped air at a minimum, and insert the top loosely.

c. Cool the assembly to 23°C or lower, then push the top down tightly with twisting motion.

d. Immerse the filled pycnometer in a water-filled constant temperature bath controlled to 25.0 ± 0.1°C, and allow the sample to come to equilibrium with the bath temperature.

e. Wipe off any excess sample, dry thoroughly, and weigh the filled pycnometer.

f. Calculate the specific gravity as follows:

$$\text{Specific gravity, } 25^{\circ}C/25^{\circ}C = \frac{A - B}{V \times 0.997}$$

Where:

A = weight of filled pycnometer, g

B = weight of empty pycnometer, g

V = volume of calibrated pycnometer, ml

0.997 = density of water at 25°C

g. Record specific gravity to the nearest 0.001 unit.

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4.5.5 **Viscosity test.** Viscosity shall be determined in accordance with the following:

a. Fill tube of No. 4 calibrated Cannon-Manning vacuum viscometer to within ± 0.5 inch of "full" line, allowing sample to run down side of tube to prevent air entrapment.

b. Condition viscometer with sample in a constant temperature bath at 25.0 ± 0.1°C for a minimum of 30 minutes.

c. Exclude all air from sample before starting run.

d. Using vacuum control capable of controlling to ± 0.1 centimeter (cm), stabilize regulator at 30 cm of mercury and attach a rubber tube from the regulator to the viscometer.

e. Read efflux time for small bulb and for large bulb.

f. Calculate viscosity as follows:

Viscosity, poise = T x C

Where: T = efflux time, seconds

C = calibration factor for bulb

g. Average readings of the bulbs and record average as the viscosity to the nearest centipoise.

4.5.6 **Iron content.** Iron content shall be determined in accordance with the following:

a. Weigh by difference 10 grams of sample to the nearest mg into a separatory funnel containing 40 ml of carbon tetrachloride.

b. Prepare wash solution of 45 parts distilled water, 45 parts denatured alcohol, and 6 parts concentrated HCl.

c. Extract the iron, by agitation for 60 ± 12 seconds, from the organic phase with four 20-ml portions of wash solution per portion. Allow at least 10 minutes for the phases to separate between extractions. Collect the washings into a beaker.

d. Evaporate the washings on a hotplate to a volume of 15 ml, add 3 ml of concentrated sulfuric acid, and reheat until all water has evaporated.

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e. Cover the beaker with a watch glass, and digest with small amounts of 50-percent hydrogen peroxide until the solution is clear.

f. Cool and dilute the digested sample with not more than 40 ml distilled water. Adjust the pH to between 6 and 7 with ammonium hydroxide. Dissolve 2 to 4 grams of ascorbic acid in the solution and transfer to a 100-ml volumetric flask. Pipet 10 ml of 1-percent weight/volume solution of 2-2'bipyridine in absolute ethanol into the volumetric flask, and dilute to volume with distilled water.

g. Mix thoroughly and allow the color to develop for at least 60 minutes.

h. Measure the absorbance of the iron-bipyridine complex on an appropriate spectrophotometer optimized at 525 nanometers. Zero the instrument using a reagent blank.

i. Prepare a standard curve of at least three points covering the range of samples, using the same procedure. Use pure iron wire in preparing the standard solutions.

j. Calculate the percent iron from the standard curve and record to the nearest 0.001 percent.

4.5.7 **Hydrolyzable chloride.** Hydrolyzable chloride shall be determined in accordance with the following:

a. Weigh a 5-g sample, to the nearest mg, into a flask.

b. Add 50 ml of 0.5N alcoholic potassium hydroxide solution, and reflux for 45 minutes. Cool the solution, add several drops of phenolphthalein, and make slightly acidic by adding concentrated nitric acid until a clear endpoint is reached, then add 2 more drops to ensure slight acidification.

c. Add 5 ml of 6N nitric acid solution.

d. Add exactly 10 ml of standard 0.02N silver nitrate solution from buret.

e. Add 5 ml of nitrobenzene and 2 ml of 40-percent ferric ammonium sulfate indicator solution.

f. Back titrate to the first reddish coloration with standard 0.01N potassium thiocyanate solution.

g. Calculate hydrolyzable chloride as follows:

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$$\text{Percent hydrolyzable chloride} = \frac{[(V1 \times N1) - (V2 \times N2)] \times 3.546}{W}$$

Where:

V1 = volume of standard silver nitrate, ml

N1 = normality of standard silver nitrate

V2 = volume of standard potassium thiocyanate, ml

N2 = normality of standard potassium thiocyanate

W = weight of sample, g

3.546 = equivalent weight of hydrolyzable chloride

h. Record hydrolyzable chloride to the nearest 0.01 percent.

4.5.8 **Infrared spectrum.** The infrared spectrum shall be determined using an infrared spectrophotometer in accordance with the following:

a. Smear a sample of the IPDI onto an infrared window.

b. Rapidly scan the smear between 5 and 15 micrometers (2000 and 650 reciprocal centimeters) without tracing the spectrum. Adjust the smear thickness so the major absorption peaks are between 0 and 30 percent transmittance and the background is between 80 and 100 percent transmittance.

c. Scan the sample from 15 to 2.5 micrometers (4000 to 650 reciprocal centimeters) using instrument settings recommended by the instrument manufacturer for a routine scan.

d. Compare the spectrum obtained with figure 1 and record any deviation.

4.6 **Inspection of packaging.** The IPDI containers and container markings shall be visually examined to verify compliance with Section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The IPDI shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.4. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National

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Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 Marking.

5.2.1 Standard marking. In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 Special marking. In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification

b. Manufacturer's name and address

c. Date of manufacture

d. Net weight of contents

e. Contract or purchase order number (exterior container only)

f. Shelf life and inspection/test date for extension (see 3.4.1). In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 12 MONTHS PROVIDED THE IPDI, UPON RETEST, CONFORMS TO THE ISOCYANATE AND MOISTURE REQUIREMENTS OF MIL-C-82813."

g. Storage conditions

h. Warning: The material contained herein is toxic and may cause extreme irritation if allowed to contact skin or eyes.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The IPDI is intended for use in the manufacture of solid propellant for the Standard Missile Mk 104 Dual Thrust Rocket Motor.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

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b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Whether first article inspection is required (see 3.1 and 6.4)

d. Material Safety Data Sheets Required (see 3.5 and 6.5)

e. Special marking, if other than as specified in 5.2.2

f. Safety precautions required (see 6.6)

6.3 **Definitions.**

6.3.1 **Batch.** A batch is that amount of curing agent made as one unit in the manufacturing process.

6.3.2 **Continuous process.** A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the curing agent is accumulated for the lot.

6.4 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 **Safety precautions.** The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the DOD FAR Supplement 28.7102. NOTE: When this specification is used as part of the description of work to be accomplished by a Government

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activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.7 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 6810-N054)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-82813	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
CURING AGENT, ISOPHORONE DIISOCYANATE

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON
c. ADDRESS (*Include Zip Code*)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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